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PROGRAM: Master of Data Science and Analytics (full time program)

RESEARCH PROPOSAL FOR MAJOR RESEARCH PROJECT (MRP)

PROJECT TITLE:

Early Sepsis Risk Prediction from Clinical Notes Using Machine Learning and Natural Language Processing.

1. INTRODUCTION AND BACKGROUND

Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection and remains a leading cause of mortality in intensive care units (ICUs) worldwide. According to the Global Burden of Disease Study, sepsis accounts for an estimated 49 million cases annually and approximately 11 million deaths globally, representing nearly 20% of all global deaths. Mortality rates in ICU settings range between 20% and 40%, increasing significantly with delayed recognition and treatment.

Early identification of sepsis is critical, as each hour of delayed antibiotic administration has been associated with increased mortality. Existing early warning systems such as the Systemic Inflammatory Response Syndrome (SIRS) criteria and Sequential Organ Failure Assessment (SOFA) score rely primarily on structured physiological and laboratory measurements. While these approaches have improved detection, they may fail to capture subtle clinical cues documented by healthcare providers in narrative notes.

Electronic Health Records (EHRs) contain large volumes of unstructured clinical text, including physician progress notes, nursing notes, and discharge summaries. These notes often include early signs of infection, clinical suspicion, and nuanced patient observations not fully represented in structured data fields. Despite this richness, unstructured clinical text remains underutilized in predictive modeling due to its complexity and variability.

Recent advances in machine learning (ML) and natural language processing (NLP), particularly transformer-based architectures such as BERT and ClinicalBERT, have demonstrated strong capabilities in extracting contextual representations from medical text. However, most sepsis prediction studies focus predominantly on structured EHR data, and comparatively fewer investigations assess the predictive power of unstructured clinical notes alone.

This project aims to investigate whether NLP-based models can accurately predict early sepsis risk using clinical notes extracted from the MIMIC-III Clinical Database hosted on PhysioNet, with sepsis defined using ICD-9 diagnostic codes.

2. PROBLEM STATEMENT

Although numerous machine learning models have been developed for sepsis prediction, the majority rely heavily on structured physiological and laboratory data. This structured-data-centric approach presents several limitations:

- Important contextual information documented in narrative notes is overlooked.
- Clinical reasoning and early suspicion of infection may not be captured in numerical features.
- Structured-only models may miss subtle linguistic indicators preceding measurable physiological deterioration.

Furthermore, while transformer-based NLP models have shown success in general clinical text classification tasks, there is limited research systematically evaluating text-only sepsis risk prediction using large-scale ICU datasets.

Without exploring the predictive utility of unstructured notes, current models risk underutilizing valuable EHR information. This study seeks to address this gap by developing and comparing traditional and transformer-based NLP models for early sepsis risk prediction using clinical notes alone.

3. RESEARCH OBJECTIVES

The primary objective of this research is to design, implement, and evaluate NLP-based models for early sepsis risk prediction using unstructured ICU clinical notes.

Specific objectives include:

- To construct a sepsis cohort from MIMIC-III using ICD-9 diagnostic codes.
- To extract and preprocess clinical notes from ICU admissions.
- To develop baseline machine learning models using TF-IDF representations.
- To fine-tune transformer-based models (e.g., ClinicalBERT) for text classification.
- To compare model performance using clinically relevant evaluation metrics.
- To assess model interpretability using explainability techniques.

4. LITERATURE REVIEW

Sepsis remains a major global health challenge, with high morbidity and mortality despite advances in intensive care medicine. Traditional scoring systems such as SOFA, qSOFA, NEWS, MEWS, SIRS, and SAPS II are widely used for sepsis screening; however, their predictive

performance is often limited due to reliance on static thresholds and structured physiological measurements.

A recent large-scale systematic review and network meta-analysis by Yadgarov et al. (2024) evaluated 73 studies encompassing over 457,000 septic patients and 256 predictive models. The pooled AUC for machine learning models was 0.825, significantly outperforming traditional scoring systems such as SOFA (AUC 0.667) and qSOFA (AUC 0.612). Neural network and decision tree models demonstrated the highest discriminative performance. However, the study highlighted substantial heterogeneity across datasets, sepsis definitions, prediction windows, and validation strategies. Importantly, only 21.9% of included studies performed external validation, and over half relied on publicly available datasets such as MIMIC.

These findings suggest that while machine learning approaches outperform traditional scoring tools, methodological variability limits generalizability and clinical translation.

Most sepsis prediction studies rely primarily on structured electronic health record (EHR) data such as vital signs and laboratory results. However, narrative clinical notes contain valuable contextual information that may precede measurable physiological deterioration.

Yan et al. (2022) conducted a systematic review examining the use of unstructured clinical text in machine learning models for sepsis prediction and early detection. Among nine included studies, most combined structured and unstructured data, and results suggested that incorporating clinical text improved predictive performance compared to structured data alone. However, the review identified substantial heterogeneity in:

- Sepsis definitions (ICD codes vs Sepsis-3 criteria).
- Prediction windows (1–48 hours before onset).
- Text preprocessing strategies.
- NLP feature representations (Bag-of-Words, TF-IDF, GloVe, ClinicalBERT).
- Model architectures.

Notably, no meta-analysis was performed due to incomparability of study designs. The review emphasized that documentation practices differ across clinical roles (e.g., nursing vs physician notes) and settings (ICU vs emergency department), which influences model performance and reproducibility.

Furthermore, many studies rely on continuous ICU monitoring signals, limiting generalizability to non-ICU settings. The review also highlighted that discharge summaries and retrospective notes may introduce temporal leakage if not carefully handled.

Despite advances in transformer-based models such as BERT (Devlin et al., 2019) and ClinicalBERT (Alsentzer et al., 2019), few studies systematically evaluate text-only prediction frameworks in ICU settings using large public datasets with clearly defined labeling strategies.

Additionally, model interpretability remains a major barrier to clinical adoption. As emphasized by Rajkomar et al. (2019), black-box models require transparency to gain clinician trust. While neural networks often achieve higher AUCs, simpler models such as decision trees may offer superior interpretability, as noted by Yadgarov et al. (2024).

This project aims to address these gaps by systematically evaluating NLP-based models for early sepsis prediction using unstructured clinical notes from MIMIC-III, employing a reproducible ICD-9 labeling strategy and a clearly defined experimental framework.

5. PROPOSED METHODOLOGY

5.1 Dataset

The study will use the MIMIC-III Clinical Database, which contains de-identified health-related data from over 40,000 ICU patients.

Link to Dataset: <https://physionet.org/content/mimiciii/1.4/>

Inclusion Criteria:

- Adult patients (age ≥ 18)
- First ICU stay only
- Availability of clinical notes within the first 24 hours of ICU admission

Sepsis Definition:

Patients will be labeled as septic if ICD-9 codes corresponding to sepsis are present in the DIAGNOSES_ICD table. Codes include:

- 995.91 (Sepsis)
- 995.92 (Severe Sepsis)
- 785.52 (Septic Shock)
- 038.xx (Septicemia group)

This ICD-based labeling approach allows for scalable and reproducible cohort construction.

5.2 Text Extraction and Preprocessing

Clinical notes will be extracted from the NOTEEVENTS table.

Processing steps include:

- Filtering notes within the first 24 hours of ICU admission.
- Lowercasing.

- Removal of protected health information placeholders.
- Tokenization.
- Stop-word handling (for baseline models)

For transformer models, tokenization will follow the pre-trained model's tokenizer specifications.

5.3 Feature Engineering

Baseline Representation:

- TF-IDF vectorization (unigrams and bigrams).
- Vocabulary size limited to top frequent tokens.

Transformer Representation:

- ClinicalBERT embeddings.
- Maximum sequence length defined (e.g., 512 tokens).
- Fine-tuning with classification head.

5.4 Model Development

Baseline Models:

- Logistic Regression.
- XGBoost classifier.

Transformer Model:

- Fine-tuned ClinicalBERT.

Training Strategy:

- Stratified 70/15/15 train-validation-test split.
- Class weights to address imbalance.
- Early stopping for transformer fine-tuning.

Optimization:

- Adam optimizer.
- Binary cross-entropy loss.

5.5 Evaluation

Performance metrics:

- ROC-AUC
- Precision-Recall AUC
- Precision
- Recall
- F1-score

Given potential class imbalance, PR-AUC will be emphasized. Bootstrapping will be used to estimate confidence intervals for performance metrics.

5.6 Explainability

- SHAP values for baseline models.
- Attention weight visualization for transformer models.

This will evaluate the interpretability and clinical plausibility of predictions.

6. ETHICAL CONSIDERATIONS

The MIMIC-III dataset is fully de-identified and publicly available to credentialed researchers through PhysioNet. All analyses will comply with the Data Use Agreement. No attempt will be made to re-identify individuals. Data will be stored securely on encrypted local systems in accordance with institutional policies.

7. EXPECTED CONTRIBUTIONS

This research is expected to:

- Evaluate the predictive value of unstructured clinical notes for early sepsis risk.
- Compare traditional ML and transformer-based NLP approaches.
- Provide insight into interpretability of text-based clinical prediction models.
- Provide a reproducible framework for text-only ICU sepsis prediction using publicly available data.

8. REFERENCES

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